LAPORAN PRAKTIKUM
INSTALASI & KONFIGURASI MAIL SERVER

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INSTALASI & KONFIGURASI MAIL SERVER PADA UBUNTU SERVER

1. Sebelum Mail Server dapat digunakan dengan nama domain, perlu menambahkan record mail pada file forward zone DNS. Buka file konfigurasi berikut

```
aisasilvana@aisasilvana:~$ sudo nano /etc/bind/zones/forward.aisasilvana
```

```
  GNU nano 7.2                                                    /etc/bind/zones/forward.aisas
$TTL    604800
@       IN      SOA     ns1.aisasilvana.local. root.aisasilvana.local. (
                        2026042201 ; Serial
                        3600       ; Refresh
                        1800       ; Retry
                        604800     ; Expire
                        86400 )    ; Negative TTL
;
@       IN      NS      ns1.aisasilvana.local.
ns1     IN      A       192.168.100.1
www     IN      A       192.168.100.1
@       IN      MX      10      mail.aisasilvana.local.
mail    IN      A       192.168.100.1
```

2. Lakukan pengecekan syntax file DNS

```
aisasilvana@aisasilvana:~$ sudo named-checkzone aisasilvana.local /etc/bind/zones/forward.aisas
zone aisasilvana.local/IN: loaded serial 2026042201
OK
aisasilvana@aisasilvana:~$ sudo systemctl restart bind9
aisasilvana@aisasilvana:~$ _
```

3. Setelah DNS mail berhasil dibuat, lanjutkan instalasi Mail Server dengan perintah:

```
aisasilvana@aisasilvana:~$ sudo apt install postfix -y
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
The following additional packages will be installed:
  libnsl2
Suggested packages:
  mail-reader postfix-cdb postfix-doc postfix-ldap postfix-lmdb postfix-mta-sts-resolver postfi
  sasl2-bin | dovecot-common ufw
The following NEW packages will be installed:
  libnsl2 postfix
0 upgraded, 2 newly installed, 0 to remove and 36 not upgraded.
Need to get 1.296 kB of archives.
After this operation, 4.321 kB of additional disk space will be used.
0% [Working]_
```

4. Setelah instalasi Postfix selesai, langkah berikutnya adalah mengatur konfigurasi utama Mail Server pada file

```
  GNU nano 7.2                                                    /etc/postfix/main.cf *
# See /usr/share/postfix/main.cf.dist for a commented version

smtpd_banner = $myhostname ESMTP $mail_name (Ubuntu)
biff = no
append_dot_mydomain = no
readme_directory = no
compatibility_level = 3.6

# TLS parameters
smtpd_tls_cert_file=/etc/ssl/certs/ssl-cert-snakeoil.pem
smtpd_tls_key_file=/etc/ssl/private/ssl-cert-snakeoil.key
smtpd_tls_security_level=may
smtp_tls_CApath=/etc/ssl/certs
smtp_tls_security_level=may
smtp_tls_session_cache_database = btree:${data_directory}/smtp_scache

smtpd_relay_restrictions = permit_mynetworks permit_sasl_authenticated defer_unauth_destination
alias_maps = hash:/etc/aliases
alias_database = hash:/etc/aliases
relayhost =
mailbox_size_limit = 0
recipient_delimiter = +

myhostname = mail.aisasilvana.local
mydomain = aisasilvana.local
myorigin = $mydomain
mydestination = $myhostname, aisasilvana.local, localhost.localdomain, localhost
mynetworks = 127.0.0.0/8 192.168.100.0/24
home_mailbox = Maildir/
inet_interfaces = all
inet_protocols = all
```

5. Agar konfigurasi baru diterapkan, restart layanan Postfix:

```
aisasilvana@aisasilvana:~$ sudo systemctl restart postfix
aisasilvana@aisasilvana:~$ sudo systemctl status postfix
● postfix.service - Postfix Mail Transport Agent
     Loaded: loaded (/usr/lib/systemd/system/postfix.service; enabled; preset: enabled)
     Active: active (exited) since Wed 2026-04-29 02:09:27 UTC; 12s ago
       Docs: man:postfix(1)
    Process: 21088 ExecStart=/bin/true (code=exited, status=0/SUCCESS)
   Main PID: 21088 (code=exited, status=0/SUCCESS)
        CPU: 5ms

Apr 29 02:09:27 aisasilvana systemd[1]: Starting postfix.service - Postfix Mail Transport Agent...
Apr 29 02:09:27 aisasilvana systemd[1]: Finished postfix.service - Postfix Mail Transport Agent.
aisasilvana@aisasilvana:~$
```

6. Setelah Postfix berhasil dikonfigurasi, langkah berikutnya adalah menginstall Dovecot sebagai layanan untuk menerima email melalui protokol POP3 dan IMAP.

```
aisasilvana@aisasilvana:~$ sudo apt install dovecot-core dovecot-imapd -y
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
The following additional packages will be installed:
  libexttextcat-2.0-0 libexttextcat-data
Suggested packages:
  dovecot-gssapi dovecot-ldap dovecot-lmtpd dovecot-managesieved dovecot-mysql dovecot-pgsql dovecot-pop3d dovecot-sieve dovecot-solr dovecot-sqlite
  dovecot-submissiond ntp ufw
The following NEW packages will be installed:
  dovecot-core dovecot-imapd libexttextcat-2.0-0 libexttextcat-data
0 upgraded, 4 newly installed, 0 to remove and 36 not upgraded.
Need to get 3.748 kB of archives.
After this operation, 12,0 MB of additional disk space will be used.
Get:1 http://id.archive.ubuntu.com/ubuntu noble/main amd64 libexttextcat-data all 3.4.7-1build1 [193 kB]
Get:2 http://id.archive.ubuntu.com/ubuntu noble/main amd64 libexttextcat-2.0-0 amd64 3.4.7-1build1 [13,3 kB]
Get:3 http://id.archive.ubuntu.com/ubuntu noble-updates/main amd64 dovecot-core amd64 1:2.3.21+dfsg1-2ubuntu6.4 [3.351 kB]
49% [3 dovecot-core 1.604 kB/3.351 kB 48%]_
```

7. Setelah instalasi selesai, cek apakah service berjalan

```
aisasilvana@aisasilvana:~$ sudo nano /etc/dovecot/dovecot.conf
```

8. Edit File Konfigurasi Mail Location

```
  GNU nano 7.2                                                    /etc/dovecot/dovecot.conf *
## Dovecot configuration file

# If you're in a hurry, see http://wiki2.dovecot.org/QuickConfiguration

# "doveconf -n" command gives a clean output of the changed settings. Use it
# instead of copy&pasting files when posting to the Dovecot mailing list.

# '#' character and everything after it is treated as comments. Extra spaces
# and tabs are ignored. If you want to use either of these explicitly, put the
# value inside quotes, eg.: key = "# char and trailing whitespace  "

# Most (but not all) settings can be overridden by different protocols and/or
# source/destination IPs by placing the settings inside sections, for example:
# protocol imap { }, local 127.0.0.1 { }, remote 10.0.0.0/8 { }

# Default values are shown for each setting, it's not required to uncomment
# those. These are exceptions to this though: No sections (e.g. namespace {})
# or plugin settings are added by default, they're listed only as examples.
# Paths are also just examples with the real defaults being based on configure
# options. The paths listed here are for configure --prefix=/usr
# --sysconfdir=/etc --localstatedir=/var

# Enable installed protocols
!include_try /usr/share/dovecot/protocols.d/*.protocol

# A comma separated list of IPs or hosts where to listen in for connections.
# "*" listens in all IPv4 interfaces, "::" listens in all IPv6 interfaces.
# If you want to specify non-default ports or anything more complex,
# edit conf.d/master.conf.
listen = *_

# Base directory where to store runtime data.
#base_dir = /var/run/dovecot/

# Name of this instance. In multi-instance setup doveadm and other commands
# can use -i <instance_name> to select which instance is used (an alternative
# to -c <config_path>). The instance name is also added to Dovecot processes
# in ps output.
#instance_name = dovecot

# Greeting message for clients.
#login_greeting = Dovecot ready.
```

```
  GNU nano 7.2                                                /etc/dovecot/conf.d/10-mail.conf *
##
## Mailbox locations and namespaces
##

# Location for users' mailboxes. The default is empty, which means that Dovecot
# tries to find the mailboxes automatically. This won't work if the user
# doesn't yet have any mail, so you should explicitly tell Dovecot the full
# location.
#
# If you're using mbox, giving a path to the INBOX file (eg. /var/mail/%u)
# isn't enough. You'll also need to tell Dovecot where the other mailboxes are
# kept. This is called the "root mail directory", and it must be the first
# path given in the mail_location setting.
#
# There are a few special variables you can use, eg.:
#
#   %u - username
#   %n - user part in user@domain, same as %u if there's no domain
#   %d - domain part in user@domain, empty if there's no domain
#   %h - home directory
#
# See doc/wiki/Variables.txt for full list. Some examples:
#
#   mail_location = maildir:~/Maildir
#   mail_location = mbox:~/mail:INBOX=/var/mail/%u
#   mail_location = mbox:/var/mail/%d/%1n/%n:INDEX=/var/indexes/%d/%1n/%n
#
# <doc/wiki/MailLocation.txt>
#
mail_location = maildir:~/Maildir_

# If you need to set multiple mailbox locations or want to change default
# namespace settings, you can do it by defining namespace sections.
#
# You can have private, shared and public namespaces. Private namespaces
```

```
  GNU nano 7.2                                                    /etc/dovecot/conf.d/10-auth.conf *
##
## Authentication processes
##

# Disable LOGIN command and all other plaintext authentications unless
# SSL/TLS is used (LOGINDISABLED capability). Note that if the remote IP
# matches the local IP (ie. you're connecting from the same computer), the
# connection is considered secure and plaintext authentication is allowed.
# See also ssl=required setting.
disable_plaintext_auth = no

# Authentication cache size (e.g. 10M). 0 means it's disabled. Note that
# bsdauth and PAM require cache_key to be set for caching to be used.
#auth_cache_size = 0
# Time to live for cached data. After TTL expires the cached record is no
# longer used, *except* if the main database lookup returns internal failure.
# We also try to handle password changes automatically: If user's previous
# authentication was successful, but this one wasn't, the cache isn't used.
```

```
  GNU nano 7.2                                                    /etc/dovecot/conf.d/10-auth.conf *

  # Do NTLM and GSS-SPNEGO authentication using Samba's winbind daemon and
  # ntlm_auth helper. <doc/wiki/Authentication/Mechanisms/Winbind.txt>
  #auth_use_winbind = no

  # Path for Samba's ntlm_auth helper binary.
  #auth_winbind_helper_path = /usr/bin/ntlm_auth

  # Time to delay before replying to failed authentications.
  #auth_failure_delay = 2 secs

  # Require a valid SSL client certificate or the authentication fails.
  #auth_ssl_require_client_cert = no

  # Take the username from client's SSL certificate, using
  # X509_NAME_get_text_by_NID() which returns the subject's DN's
  # CommonName.
  #auth_ssl_username_from_cert = no

  # Space separated list of wanted authentication mechanisms:
  #   plain login digest-md5 cram-md5 ntlm rpa apop anonymous gssapi otp
  #   gss-spnego
  # NOTE: See also disable_plaintext_auth setting.
  auth_mechanisms = plain login

  ##
  ## Password and user databases
  ##

  #
```

```
  GNU nano 7.2                                                    /etc/dovecot/conf.d/10-master.conf *
  #process_limit = 1024
}

service auth {
  # auth_socket_path points to this userdb socket by default. It's typically
  # used by dovecot-lda, doveadm, possibly imap process, etc. Users that have
  # full permissions to this socket are able to get a list of all usernames and
  # get the results of everyone's userdb lookups.
  #
  # The default 0666 mode allows anyone to connect to the socket, but the
  # userdb lookups will succeed only if the userdb returns an "uid" field that
  # matches the caller process's UID. Also if caller's uid or gid matches the
  # socket's uid or gid the lookup succeeds. Anything else causes a failure.
  #
  # To give the caller full permissions to lookup all users, set the mode to
  # something else than 0666 and Dovecot lets the kernel enforce the
  # permissions (e.g. 0777 allows everyone full permissions).
  unix_listener auth-userdb {
    #mode = 0666
    #user =
    #group =
  }

  # Postfix smtp-auth
  unix_listener /var/spool/postfix/private/auth {
  mode = 0666
  user = postfix
  group = postfix
 }

  # Auth process is run as this user.
  #user = $default_internal_user
}

service auth-worker {
  # Auth worker process is run as root by default, so that it can access
  # /etc/shadow. If this isn't necessary, the user should be changed to
  # $default_internal_user.
  #user = root
}
```

9. Agar konfigurasi diterapkan, restart layanan:

```
aisasilvana@aisasilvana:~$ sudo systemctl restart dovecot
aisasilvana@aisasilvana:~$ sudo systemctl restart postfix
aisasilvana@aisasilvana:~$ sudo systemctl status dovecot
● dovecot.service - Dovecot IMAP/POP3 email server
     Loaded: loaded (/usr/lib/systemd/system/dovecot.service; enabled; preset: enabled)
     Active: active (running) since Wed 2026-04-29 02:26:21 UTC; 14s ago
       Docs: man:dovecot(1)
             https://doc.dovecot.org/
   Main PID: 24494 (dovecot)
     Status: "v2.3.21 (47349e2482) running"
      Tasks: 4 (limit: 4544)
     Memory: 3.2M (peak: 3.5M)
        CPU: 53ms
     CGroup: /system.slice/dovecot.service
             ├─24494 /usr/sbin/dovecot -F
             ├─24495 dovecot/anvil
             ├─24496 dovecot/log
             └─24497 dovecot/config

Apr 29 02:26:21 aisasilvana systemd[1]: Starting dovecot.service - Dovecot IMAP/POP3 email serv
Apr 29 02:26:21 aisasilvana dovecot[24494]: master: Dovecot v2.3.21 (47349e2482) starting up fo
Apr 29 02:26:21 aisasilvana systemd[1]: Started dovecot.service - Dovecot IMAP/POP3 email serve
aisasilvana@aisasilvana:~$ _
```

10. Setelah Postfix dan Dovecot berhasil dikonfigurasi, langkah berikutnya adalah membuat user baru yang akan digunakan sebagai akun email pada Mail Server.

```
aisasilvana@aisasilvana:~$ sudo adduser user1
info: Adding user `user1' ...
info: Selecting UID/GID from range 1000 to 59999 ...
info: Adding new group `user1' (1002) ...
info: Adding new user `user1' (1002) with group `user1 (1002)' ...
info: Creating home directory `/home/user1' ...
info: Copying files from `/etc/skel' ...
New password:
Retype new password:
passwd: password updated successfully
Changing the user information for user1
Enter the new value, or press ENTER for the default
        Full Name []: aisasilvana
        Room Number []:
        Work Phone []:
        Home Phone []:
        Other []:
Is the information correct? [Y/n] y
info: Adding new user `user1' to supplemental / extra groups `users' ...
info: Adding user `user1' to group `users' ...
```

```
aisasilvana@aisasilvana:~$ sudo adduser user2
info: Adding user `user2' ...
info: Selecting UID/GID from range 1000 to 59999 ...
info: Adding new group `user2' (1003) ...
info: Adding new user `user2' (1003) with group `user2 (1003)' ...
info: Creating home directory `/home/user2' ...
info: Copying files from `/etc/skel' ...
New password:
Retype new password:
passwd: password updated successfully
Changing the user information for user2
Enter the new value, or press ENTER for the default
        Full Name []: silvana
        Room Number []:
        Work Phone []:
        Home Phone []:
        Other []:
Is the information correct? [Y/n] y
info: Adding new user `user2' to supplemental / extra groups `users' ...
info: Adding user `user2' to group `users' ...
aisasilvana@aisasilvana:~$
```

11. Setelah user email berhasil dibuat, langkah berikutnya adalah menguji koneksi Mail Server menggunakan telnet.

```
aisasilvana@aisasilvana:~$ telnet mail.aisasilvana.local 25
Trying 192.168.100.1...
Connected to mail.aisasilvana.local.
Escape character is '^]'.
220 mail.aisasilvana.local ESMTP Postfix (Ubuntu)
EHLO aisasilvana.local
250-mail.aisasilvana.local
250-PIPELINING
250-SIZE 10240000
250-VRFY
250-ETRN
250-STARTTLS
250-ENHANCEDSTATUSCODES
250-8BITMIME
250-DSN
250-SMTPUTF8
250 CHUNKING
MAIL FROM: user1@aisasilvana.local
250 2.1.0 Ok
RCPT TO: user2@aisasilvana.local
250 2.1.5 Ok
DATA
354 End data with <CR><LF>.<CR><LF>
SUBJECT: tEST eMAIL

hALO USER2, INI EMAIL DARI USER1!
.
250 2.0.0 Ok: queued as 6F6D01052F3
QUIT
221 2.0.0 Bye
Connection closed by foreign host.
aisasilvana@aisasilvana:~$
```

12. Setelah pengujian pengiriman email menggunakan telnet berhasil, langkah berikutnya adalah memastikan email benar-benar masuk ke user tujuan.

```
aisasilvana@aisasilvana:~$ sudo cat /home/user2/Maildir/new/1777430509.Vfc00Ia35aaM33400.aisasi
Return-Path: <user1@aisasilvana.local>
X-Original-To: user2@aisasilvana.local
Delivered-To: user2@aisasilvana.local
Received: from aisasilvana.local (ns1.aisasilvana.local [192.168.100.1])
        by mail.aisasilvana.local (Postfix) with ESMTP id 6F6D01052F3
        for <user2@aisasilvana.local>; Wed, 29 Apr 2026 02:40:24 +0000 (UTC)
SUBJECT: tEST eMAIL
Message-Id: <20260429024052.6F6D01052F3@mail.aisasilvana.local>
Date: Wed, 29 Apr 2026 02:40:24 +0000 (UTC)
From: user1@aisasilvana.local

hALO USER2, INI EMAIL DARI USER1!
aisasilvana@aisasilvana:~$ _
```

13. Selesai!